

ActInf Livestream #017.2 ~ Information flow in context-dependent hierarch

Livestream | Information flow in context-dependent hierarchical Bayesian inference | 1:34:42

hello

welcome to active inference lab and

active inference live stream 17.2

it's march 16th 2021

welcome to the active inference lab

everyone we're a participatory

online lab that is communicating

learning and practicing applied active

inference

you can find us at our links on this

slide

this is recorded in an archived live

stream so please provide us with

feedback

so that we can be improving on our work

all backgrounds and perspectives are

welcome here

and we'll be following video etiquette

for live streams

at this shortened link you'll find our

calendar for active inference live

streams for 2021 and here we are

in the follow up to 17.1

in the same paper and uh definitely

recommended to check out 17.0 and 17.1

if you haven't but this will hopefully

be a cool discussion either way

and you can also see the future

live streams cool well today in

17.2 we are going to be just continuing

to discuss

and learn this really interesting paper

and

it will be fun to speak with whoever

joins or doesn't for this stream and

also hopefully to
get some questions from the live chat
but we're just going to introduce
ourselves go over some general
questions and i think uh seeing as it's
a dot two
and we had a lot of the background and
dot zero and then we had um
chris fields the author come on for dot
one
let's just think about where we can take
this fun
discussion okay so here
um especially for our first time
participants we can introduce ourselves
and just warm up so first we can just
introduce ourselves
i'm daniel i'm in northern california
and i'll pass it to
stephen hello there i'm stephen i'm in
toronto canada
and yeah i'm interested in a number of
areas
that relate to active inference
particularly those that can inform
community-based development and the way
that we
make meaning in complex contexts
and i'll pass it over to
is that alex there dean
dean sorry
hi um i'm dean uh i don't know how to
pass things yet
i'm pretty new to this um i'm in
calgary so hi steven in toronto
and uh i decided to s
to step into this because of
uh reach out to daniel did on

researchgate and
a follow-up and then the opportunity to
look at last week's 17.1
presentation and for about two hours i
was completely
absorbed in the conversation and thought
this is really interesting so i
took the next step and decided that i
would try and uh
integrate myself more with what's going
on here
cool well let's maybe learn more about
your
perspective as we walk through some of
these warm-up questions
and just go through at whatever
pace we want just sort of the topics in
the paper as well so the warm-up
questions are what is something you're
excited about today
what's something you liked or remembered
about the paper or
also the discussion 17.0 or one
and what's something you're wondering
about or would like to have resolved
so maybe since we all listened to it and
hopefully those listening will have two
what was something about 17.1 that sort
of
stood out in people's mind or something
that changed how they thought about
active inference or how they thought
about
something that wasn't active inference
yep stephen
yeah well one of the things that was
quite interesting was that
there are other ways of

people looking at the math side of it
which
some of you know the math itself is
normally quite a challenging area
to get your head down but to realize
there's also some other thoughts
happening
at some foundational level so that um
was interesting i didn't know about that
particularly
um or to the extent that was happening
so that that was particularly one area
and
that tied into this idea of quantum
contextuality
which i hadn't heard of actually and
really
was very interesting
dean what do you think um i guess the
biggest thing for me was and i and i use
terms that may not you you might have to
unpack them but
one of the things that i was really
interested in listening to was the fact
that um
that pers the perspective piece and how
you
every time you're observing something
the perspective that you hold
um you may not be aware of
how important that is but the fact that
chris sort of brought that out again and
again
um well that really that really
resonated with me because it's something
that i've been working with
with people for a long time cool
so a few thoughts that i had from 17.1

and also what you just said
so this quantum contextuality and then
the part that was really novel and
exciting to me
was thinking about quantum not as just
simply about
electrons and protons and okay you got
strong force weak force
electrons protons maxwell's you know
just throw the regular physics particle
physics or something
and then it's just equations that have
different efficacy
over different spatial scales that's one
presentation of quantum
and so then it's like well if you were
that size or there's something about
uh the system about that specific system
or that size of system that makes it
quote quantum
and this just expanded the definition of
quantum to me to this kind of
measurement communication
encryption encoding and ultimately also
bidirectional meaning and semantic
information
all the way down so that was a very
different way to think about
the relationship of like the molecule to
the electron orbital
it's like it's quantum it's not the way
that the electrons would be behaving by
themselves but maybe that's not just
because they're
electrons maybe that's something about
structure
in general so that was pretty cool and
then also you mentioned

and chris did dean observation is always
coming from a perspective
and that was something that we talked a
lot about with the projective
consciousness model
from an earlier discussion and so that
was saying
yes um you get all these cybernetic
functions
and all these functional attributes of
consciousness
from a projective geometry where the
observer is at this the center at this
actual special point
and then things are projecting out in a
certain way
so that's like kind of physical analog
it's geometric
and then um stephen you said there was a
different sort of math that they were
presenting
they didn't use projection of geometry
this was from like a different branch of
the game with the category theory and
with these other
theories that we just were barely
scratching the surface on
so that you know what does that mean
that those different branches can re-uh
integrate in a certain way and then
potentially not just through
an equation but through a process
theory so that was some interesting
stuff
any other thoughts or i mean really yeah
we're just we're just
yeah go ahead just now that's some
really that's really interesting

i'm just letting it sit because it takes
this idea of the perspective and this
idea that you know the idea that this
quantum idea can be much more across
scale and he did talk about i think he
called it quantum noise or
there was the use of noise which then
gives a rationale
i think for how this quantum processes
could permeate through everything
even if you don't see quantum
transitions in the simple
sense like you said with electrons and
stuff then there's
this this is permeating and that that
then comes
to that idea of the the the
contextuality
as well as perspective when we think we
take a bit different perspective
in a very okay i can see i've moved
from here to here and i've taken a
perspective but perspective
at this kind of phenomenological deep
level
is kind of happening happening at um
um all scales like the the context is
changing
um beyond what we can take a perspective
on if that makes sense except we use the
word perspective because
that's what we can get a handle on but
it's it's changing in so many ways of
which some of which we just can't
know so i thought that was really
interesting
um and maybe that's something that our
biology

is knowing but a deeper level you know
it may be these
nested levels there's ways of which that
sort of knowing is being
harvested once for a better term um
so yeah thanks
the body knowing which is something we
hear about and something you just said
it's almost like well if the body knows
through action like the tendon knows to
do this because of its biomechanical
properties but that's it's
how the knowledge it's not a turing
computer it's not doing
some trigonometry but it involves angles
but it's because of
how it's embodied so that's a it's an
it's not the knowledge on a hard drive
so that's sort of why some of these
information science approaches are
probably
on the cutting edge of what information
means because
we're having to think about how there is
information like
in the body of the insect that helps it
walk or
in the nervous system of the insect
there's some pattern generators for
example that
that's where the knowledge of walking is
so it can't do every walking pattern
but it is like this attractor for
walking
and then that's where the knowledge is
but it's not all knowledge and it's not
a text file
or script that is implemented on

how to walk or not even really an
algorithm

so it's kind of like

good and also yeah knowing is on the
inference

side and then what is the other side of
that

with action

[Laughter]

so i use sometimes use

the frame of of prisms and prismatic

and you got energy in and then you got a
rainbow on the other side daniel are you
kind of talking about

seeing both at once

like knowing what both are are doing i

mean it's going to go

through this triangle right but i mean

just being aware

that what what what we see on one side

is not necessarily what we see on the

other side and that doesn't make

one more important than the other it's

what's important is knowing

how things are are

looking to the person who's got this

perspective now

okay i'll try to address it yeah i seem

to remember

shauna saying the word prismatic or

chromatic or something about

yeah i use that all the time yeah okay

it probably has an interesting

technical definition so it's interesting

just that it's like a vivid sensory

representation of all the opportunities

all the possibilities all the colors the

rainbow

and even beyond with the electromagnetic spectrum

greetings blue welcome um

and then there's like um yes the prism focusing different rays of light or just optics like

the rays of light are focusing at the center point

and then um that's where the focus is that's

where the regime of photons is and then that's what enables the function

um greetings blue do you want to say hi and give any

um thoughts while i just resize some stuff yeah sure

good morning everyone my name is blue knight i am an

independent research consultant based out of new mexico

and yeah this paper was i don't know it was like really difficult

to work through i thought and also really well written

for like i'm not like want to sit and read like math

lemma this and theorem that and like that's always like not

my thing i mean i always find it just difficult to digest i like to do math as problems but not read it as like reading material

but i thought that for like the amount of information that was contained in this paper it was like really

well written and easy to relate to

um i did think that it could have used some additional like practical

examples of stuff but then like
considering the length of the paper i
was like man if they like put practical
examples for all of these different
things it would have been like a 70 page
paper

so um those are my thoughts on the paper
and it was great having chris
here with us uh last week to answer all
of our questions
and definitely something i um really
enjoyed

so cool

um yeah i wonder if we can
keep that question about examples of
this

in mind as we maybe look through
the keywords or look through the
roadmap let's just re-acquaint with the
paper

so this is the information flow in
context-dependent hierarchical
bayesian inference by fields and
glazebrook one of their
papers and collaborations together and
they basically lay out that there's two
goals they're kind of a twin purpose
with this paper

so one piece is related to
what stephen was bringing up earlier
with

taking this quantum contextuality
approach

and then generalizing it in a scale-free
aka multi-scale way

using techniques that are scale free
apparently such as two spaces and
channel theory

although also say like a linear regression is scale free there's no a priori scale for a linear regression it could be over little things it could be over big things so in a way just saying scale free is almost like saying we made it about how the modeler approaches the system so that we didn't get tied down in attributes of the system that we don't want to be um making skill-free and then the second side of this scale-free approach to contextuality is they're going to demonstrate a connection between that and a lot of recent work in bayesian statistics and hierarchical bayesian models which is also where the most direct connection to active inference comes into play because we've also seen hierarchical bayesian models as implementing active inference and then that's being tied to something that's at a different or a higher level of mathematical generality which potentially could allow an active inference model that you make for this specific task if you make it within that framework you might be able to lift it out and put it somewhere totally different like you might be able to use it to describe systems and fit parameters or generate data that's something that

not all models can do
but there's probably other modeling
maneuvers that will be possible
and so it's kind of like if you have
gaussian assumptions certain things and
statistics work really well together
like you know certain relationships
between multiplying things and adding
variances
so this is kind of like a higher level
statistical
um relationship
that and i think that has to do with the
matu the commuting of the cones
or the cocons but who knows
um okay on the road map were there any
sections that like sprung
to mind or anything somebody wants to
say on this
all right let's go to why it matters and
think about the
examples so yeah stephen
um yeah okay and good points you just
made
um i think that the um
that example that was just given
actually about the prism
i think it was a good one i think that
might even itself be something that
could even be a a topic of discussions i
think that opens up things but it also
uh
it gets some other stuff because one
thing that's going on here is
the the bayesian inference you're kind
of inside
the the situation that's at play
and what i think that if we're going to

say why this matters i think it might
relate
interestingly to a paper that's going to
be actually coming
up over the next week which talks about
um the way that we understand the global
workspace
or theory of the brain and the idea that
we have this phenomenological
consciousness which is somehow
attuned to the flux of sensory data
and then we have what we're actually
often modeling which is what we're
able to consciously be aware of
so this flux that's at the you know
which
sort of it starts to get into the
markovian monism
theories of of um fristen
this seems to sort of give another way
to go from the flux
up uh maybe this you know
in the way that david bone talks about
the um implicit order
of nature and um and i got a message
from tim as well
just through saying about to think that
you know we've got this idea of quantum
noise and classical noise
you know and you know this non-linear
dynamical systems are kind of working
with what could be mean as noise
or changes in noise so um
so i think that that part of the
that part of the the story is um
is sort of important across all of
active imprints in a way
because at some level there has to be a

way to
extract meaning from
uncertainty and so it seems that this
gives some other way into that
blue or dean
here's a thought okay go ahead dean i'm
just going to say that
i so back to the perspective thing
um so one of the hardest things that
when i was working with
with people trying to find themselves in
in new situations was you
you you're either inside of the box or
you're outside of it and i like the word
that stephen used their
extraction because i think that that
i think what this work does is it maybe
enables that
extraction piece a little easier if
people
know what you're talking about in terms
of stepping back
we have a tendency to step into things
and analyze things from a from an
inside out perspective and i think
how this how this could potentially work
is as a way of of gaining
a sense of removal which is actually
advantageous at times i mean if you're
looking at
something that's complex it's it's very
difficult sometimes not to get
caught up in the in the moment i was
speaking
just before the live stream started
about
white water rafting when you're when
you're going down the river and you're

in a class

three uh rapid you're thinking about

that you're not thinking about

the perspective of being up on the cliff

watching the raft go down the river i

think this allows people

to get outside of that moment i think

that's my

one of my hopefully why this matters

contributions

even well just sort of

with with that i think in some ways i i

agree what you're saying but i would

also

say what's interesting is maybe it

allows you to remove yourself

more than we normally can because but

also show that actually when you think

you're immersed in it

you aren't as immersed as you think you

are right because there's these other

levels of

immersion um which are you're

beyond your conscious awareness and

there's probably even other levels which

even beyond your phenomenological and

biological

systems ability to tap into so there's a

maybe it's both ends if that makes sense

you go you can go further in if the word

in is meant that way and

further out if the word out is meant

that way

so on this why it matters um

here's one thought and then anyone else

or i could go for a second so

stephen when you said extraction um and

i know you have a background in

chemistry

so that's sort of like it's separating

into the different fractions

so it's it's um not just extracting like

you know the lithium from the ore it's

actually

like we can have a discussion what are

the affordances what is on the table

what are we just going to just say is

our option space for policy

and then what are our preferences and

what kind of uncertainties are we each

bringing to the table from our

perspective

so it is um not like we're extracting

the essence of the conflict and then

we're gonna

fix the essence or the reduction of the

conflict or one dimension of the

conflict perceived

and then that will propagate through the

other

aspects of the situation it's kind of

like we're gonna come with uncertainty

to a minimal model

and then if one person has one narrative

for the policy and somebody else has a

different narrative

that's off the table so that is a

way where people can bring their

different ways of seeing

a common model differently like in the

quantum measurement case

if both people are measuring the quantum

particle and they're both fine with it

that's one type of experiment steven

yeah i like the point you picked up

there on extraction it was interesting

because

uh that that sort of came up so strongly

and i think um

the other thing that's interesting is

the idea even in chemistry is the idea

that still you extract

something right it's quite thin like but

the idea that you're extracting

something from noise

because in chemistry it's normally like

basically you you calculate everything

down and then say what is

negligible what can you say is beyond

concern so you just say it's beyond

worries you know it's it's within

tolerance so to speak and then you just

work within that whereas in this case

it's like well

all that noise so to speak is what we're

extracting

the information from it it feels to us

like the stuff's out there and we're

getting

thinness so to speak but we're actually

getting

signals and supposed extracting enough

variational free energy changes in the

noise

to then perceive it as you know

knowledge but it actually it has to

start out as

fluctuations in i don't know if the word

noise is maybe the wrong word but

certainly fluctuations in non-linear

dynamics

so that's that's that changes a lot of

things about how we understand knowing

the world

interesting here's kind of another point
on the signal to noise
so we're thinking in the shannon
information aka shannon disinfo theory
nothing against it framework it's about
the telegraph
and the wire and getting the exact
recapitulation of the signal on one end
of the wire as the other end of the wire
that was the second paragraph of the
1948 paper by shannon
we talked about it in the dot zero video
he said the question of communication is
about perfectly reconstituting a signal
and so in that framework if you have if
you're doing a chemical synthesis
and there's one product you want then
you can calculate like the yield of that
product
and then similarly if there's one thing
that you know that you want you know you
can calculate
really exact statistics like information
um
and then once you open it up to the
semantics
um then this the noise has a different
perspective just like you are
highlighting cure i think there's a lot
of
richness in what you already said but
yes noise is richer than just
decay in the copper wire because we're
not trying to
recapitulate anything in the person's
mind it's not a bit wise transfer
through speech
it's actually like the differences in

the way that we speak are
conveying meaning so it's not just error
from some perceived
um pattern of speech it's like just
the meaning that the person is conveying
through their action so that i think is
also
something that really matters when we
think about differences in how people
are going to be learning or
communicating or something like that
stephen yeah and
the the interesting thing is that while
to some extent you heat up a liquid or
something it expands and you sort of
the energy put in shows itself as the
work you particularly see say when water
freezes
but there's this whole question of
entropy
and how it somehow tied to some
rearrangement
which is not shown as work in the system
you know
that you try and heat up ice it hits a
point there's something going on there
and we can't quite
know what it is and it's energy hungry
in some instances or energy yielding
but um that that process is beyond
our direct
perception so you just have to build it
in with gibbs free energy
and the entropy and enthalpy terms
to see what will happen with reactions
so that
that actually then actually ties into
fristen's work

um in in actually it's the sort of
foundation of his work in a way
so i i'm not sure how entropy
was named in this paper maybe i don't
think that's directly part of this math
this is another way to come in without
maybe coming in with the word entropy
but um it's interesting
thanks dean i just
again just try to draw from examples
when i would talk about this
with people and there's a privilege
always given over to gap
closing like we want to close the gap we
want to understand we want to find the
signal within the noise
and i think what this does is it sort of
gives over a certain amount of
credibility just a gap respecting
just leaving the thingness out of it for
a bit
so that you can actually take in more
information back to what stephen was
and what you're both alluding to how do
how do we
how do we give ourselves um
cut ourselves a little bit of slack by
saying gap respecting is just as
important
as gap closing and if we give ourselves
that it's a kind of a privilege to do
that because it's twice as much effort
as just gap closing i think the benefits
can be really
quite profound nice
point thank you so sarah oh stephen
do you have anything to add here well
just one last point on that is that

could then
where you start to get into abductive
growth based
knowing um where is maybe
um we're used to because humans have got
this capacity for deductive
and rapid you know exclusion of what's
out there
an inductive trying to narrow the gap
between us and a goal
but when we're growing when we think
about sometimes i need to just sit back
and heal
you know there's certain situations in
the world which i can never process
psychologically from a i can't close the
gap on certain things right i just have
to sit back
and heal right because they're hard and
so that
that same piece you're talking about
there then starts to tap into well
that's the main game in town for
organisms
yeah the gaming town of closing gaps and
is is um is a nice add-on that we have
but it isn't the main game and
blue
so that just brings up to mine's like
time as a gap closer right like when you
just sit back and heal
and and ultimately like time as a gap
closer like if you think about like the
time
value of money and i mean like time is
ultimately the thing that will you know
close the gap on
all prediction and and like we don't

really see that in
the models or in the math at all um it'd
be interesting to see like the temporal
dimension and how that plays in or
people have thought about that or i mean
time like can it flow forward and
backward
and i think that was something that was
brought up maybe in the last live stream
that's just interesting yeah nice dean
just just one last thing i i i
completely agree with you because
again the privilege goes on on to time
on task
instead of task or or task on time
instead of just seeing time in and of
itself i
have this crazy theory that that time is
learning
not just a way of measuring learning
and i don't you can't get there unless
you can get off that
treadmill so thank you that's a great
point
and dean what you said about learning as
time it's like the difference between
kairos and kronos
so kronos the chronometer the decimal
time and it's about the wall
time the clock time and then action is
happening
in a space of meaning of act it's time
for dinner and that's an action
affordance and so that is something
that's we can bring in the uncultured
component whereas the other approach is
no the wall time is just
simply how it is and we're going to like

pro trustees

shrink expand use heuristics

use coarse graining to get around this

sort of fundamental

um linearizing from our perspective

just at the scale we're at which makes

it not scale-free

which is why perhaps quantum only

applies to certain so-called spatial

levels of what

we measure steven

yeah sorry give me a sec i had it and

then i just went for a sec there so

let me ask sarah's question because it's

really nice point in chat

so sarah wrote um could we look at chew

space

slide and get a broader perspectival

view to maximize

signal to noise sort of

speak so maybe i didn't perfectly read

it but i think

it was two spaces can we enter into here

a little bit and connect that to some of

the bigger

discussion points and maybe even the

other um

useful points so stephen you want to

give a thought or is there anyone who

wants to give a thought on the two space

piece here

yeah does it this directly i'm not sure

exactly how but

um this gives a way that um things can

transform from one space to another

so it gives you that capacity i mean

sort of keep it tightness to the the

idea of time i mean with

and i know this paper is not specific
active inference with active influence
you know
without time you're not able to make
inferences from
noise so to speak or changes in
information content
in random fluctuations you only get that
because you've got time because then you
can tell if there's been a fluctuation
so one snapshot can't do that now
this seems to maybe give away to okay
you've now got a shape of the
information space
what happens once it has a shape
um now but beyond that it's uh
i'm not entirely sure
yeah um blue anything you'd say on this
um topic otherwise i think walking
through
kind of just the way they laid it out in
the definitions could be helpful in some
of the examples they gave
because like steven you you mentioned
this morph
ism right and it seemed like there's the
true space
and that's like a matrix but one that
can be various
kinds of things it's a matrix but just
like we can represent
a network with a matrix so we can think
about like the space of all the networks
all the social networks of five people
it's like a five by five matrix and ones
and zeros or weighted numbers
so that's the matrix representation of
the network

and then that is one type of meaning of
the matrix
but also this can describe probability
and it can describe
programming languages bayesian
networks like if each cell had
information about a conditional
dependence or not
so that's like a a chew space is the
space of the possible
of these things and then
this is the part where maybe we'll just
say it's beyond the experience on this
conversation but
the wikipedia has like understood
statically
it's a relation and then understood
dynamically and then it has another
definition so it gets interesting has a
static or a dynamic
interpretation so i think it's
yeah unless anyone in the um chat wants
to jump in and help or somebody with
expertise wants to come on later
um how can we connect that to some of
these use
cases blue or that on anything else go
ahead
so i think like the um the dynamic
aspect of the two space and the the two
morphisms is that like
you know any um if you have like a set
a matrix right so if you have a matrix
and
it's like any series of transformations
that can be done to form
a new matrix so that that in my mind is
like the dynamism

of it all like if i take you know set a
and i do
this that this that this that in this
specific order
so again like here here's like the
temporal like we're alluding to like
time as a variable or
time as a gap closer but like the
temporal sequence is important because
you know if you do
you know if you multiply before you
subtract or if you get your order of
operations wrong it's going to be
different right so the temporal order of
things
is the the morphism to to make the the
next
set
thanks for that another interesting
sentence
is that um it's like the true transform
or the morphism
is a pair it kind of comes into
existence
as two spaces and a mapping between them
or a map
between them and so that is what ties it
really broadly to functions and
computation
and these sort of transformation
transmutation or input output seeming
communication seeming processes
measurement like
processes because it's about sum space
and some other space
and so when we think about the active
inference model
and then we have hidden states that

inference is being done on
and then we have observations and it's
almost like
there's a matrix that connects the two
of them i'm not saying that that's going
to be
all at this level of two space but it's
kind of like
there's a mapping where you can run the
model from the sense data
to the state and do inference or you can
generate from the state
likely sensory data so that is a
bi-directional it's like a two-way
bayesian machine because it can take in
observations and update
expectations of hidden state variables
or it can generate
so that generative and receptive element
of bayesian computation is i think just
it's like whatever is the topic that
includes that
and that network matrix and a few other
things
that's kind of where this topic can go
which is why i think it's a
really rich area and that's why the
papers are written in
2020 and that's that's
yeah stephen and then dean
i think this this idea of moving between
spaces is also really
valuable because as you know i'm very
interested in
um understanding how we can look at
awareness
and sort of indigeneity is very much
about awareness

and a lot of the active infant work work
is tends to be at whatness what's this
what's that how can we understand what
this is understood to do
and think about but this when you start
to get to this broader
um understanding of how we can make
meaning
um and taking that idea a very broad
level
is the ability to transform between
spaces and engage
with spaces and that might extend up to
our conscious awareness space
but some of these spaces are well below
the level of consciousness you know down
to the level of cells you know but
it's it seems to have and play out the
importance of
spatial transformation over whatness
thank you steven dean yeah
what i took away from this was that
when we when we discriminate when we
separate
um this sort of gives you a sense of
why we might do that um i think of
some of the stuff that we looked at um
in some of the work that i did we would
ask people to
watch something maybe a youtube clip
and instead of trying to take all the
information in as
is is classically recorded through notes
we'd ask them to columnize and row
in under three headings tools
and rules and pools now that was the
mnemonic but
pools basically where the gradient is

what's

aggregating and so we'd ask people to
classify

something and what people found was
something could fit into

two categories something could be a tool

and a rule sometimes it could go across

all three it could be a tool

and a rule and a pool like there was a
way of

being able to see um relationship

and i think that's what this two spaces

does it says

we can discriminate we can separate but
sometimes

things fit into multiple categories and
that's

that's something we need to be aware of

our minds are able to sort of

pull apart and then reorganize that's

what i took away from that

nice very nice dean and i think that's

sort of reflected by

this cone and potentially even your
prism

topic because it's like yes there's a
unity of

c in this case but it is going to be

manifest in these different

spaces and this is a really informative
slide

so it's talking again about these pair
of two

spaces that are related through a
transform

the transform a to b can't and this is

from section or example 2.7 in the paper

the transform a to b in this case can be

viewed as transporting the information
encoded in valid flow formulas from a to
b and can thus be thought of informally
as a channel from a to b implicitly
providing a sense of spatial
and or temporal separation between a and
b so i thought like
there's the maps of meaning and so
there's my
internal map just so to speak just using
it instrumentally and then
there's person b and then if
we're playing 20 questions it's like
okay is the house bigger or smaller than
a bread box
there's some semantic mapping internal
to me and then there's a channel
with this with the sense data and then
there's a semantic flow related to
whether they're going to give me a yes
no
which is like chris said that's quanta
because we're quantizing the response
is it bigger or not we're making a
measurement of the response now
they could be lying or they could be
wrong or
they could you know we couldn't hear
them or something but it's a binary
question and then
there's a semantic flow and then there's
a back and forth
potentially otherwise you're talking
about different things and so
if somebody is thinking maybe i was
thinking of a little toy house
smaller than a bread box so if you have
a different

frame of reference then even the
communication is not going to
work because the holograph is going to
be read
too differently by the people even with
the same information
so that's like the separation of
cognition across
participants in a conversation and then
that's happening
implicitly through time but then it
seems like also
things can be just more abstractly
described as being separated through
time
and so something that's separated in
time but not in space
is like memory which is where this whole
discussion of memory comes into play
because it's it's a communication
through time at that spot like a time
capsule
it's like 50-year message out
you debate whether you're the absolute
position of the time capsule is the same
but it's like relative to it
it's a message through time blue
so it's funny that you bring up a time
capsule like i've been really thinking
about um
you know at scale carrying information
forward
and backward like from you know cells to
person
to like a group of people and and it's
like you know
when we what do we carry forward and
when we like

used to make time capsules like every 10 years i don't know if we still do but then we would dig them up and like you know dig them up every 20 or 30 years or whatever and like what do we put into a time capsule and does that like effectively capture like the culture of the society at that time like a cd or like you know these different things that go into the time capsules like this is actually like what we're carrying forward and sometimes it's like dumb stuff like i don't know it's just funny to think about what goes forward right in time and and why that and like how the selection process works and if it's like a systemic or generalized thing or um i don't know it's just something that i've been thinking about how that is transmitted across time great question and very prescient of course with so much digital information and a potentially fluctuating cost of storage wildly cheap but then wildly expensive and that could lead to some very unfortunate information flows evolution has sort of an answer to that which is like what's passed forward is what is fit to its niche not biggest bench press strongest fittest but fit to the niche adapt skilled

to the niche so we don't need to go too
semantic on that but i think the term
actually does
fit and where evolution by natural
selection
as well as active inference come into
play
is by formalizing a process theory
you kind of step away or at least you
can maybe get some distance or some
model ability of these complex processes
so in evolution the process theory is
basically
you have heritable variation in
phenotype that's found
in a population where there's heritable
variation in fitness
and if those are all linked up in the
right way at least locally
that's the way that the phenotype and
the population moves and so active
inference
it's similar but kind of like with
information just saying okay sense data
comes in
action goes out from a generative model
into the niche
back into the sense something happens
with the generative model integrates the
information
engages in policy so it's like if
somebody listens to a lecture and they
only write the notes
of five sentences and then they pass
that off the process theory
explained why there was an information
bottleneck there because we just modeled
the situation and we don't need to go

into oh it's tragic there was only five sentences or there was more than you know it was five deep sentences though or something like that we just say what it was that was the actual transmission and then that is a starting point for the kinds of value-oriented analyses that you might want to do afterwards dean yeah i'm just curious what other people's thinking is on this because typically between the inference and the model we not necessarily the people here but oftentimes what's parked in between those two things is a plan and what you just described is not a plan so i'm curious what other people think is between the inference and the model i have my ideas but i'm because i'm trying to get to know you folks i'd like to know what you guys think is between the entrance and the model stephen yeah well this is that's the six million dollar question they um i i think that the the the introduction of the quantum approach at certain scales does yield something very plausible and and and chris's work with mike levin would tie into that because he does a lot of work with bioelectrics

on cells membranes so i could see
that the this question where it's more
directly attributed to to
quantum fluctuations and what that means
you know at the level of the cell and
then the bio
electrical or the other chemical
electrics or the bioelectrics of the
membrane
and then you the next level that i can
kind of see it more plausible is brain
waves you know
and you've got this overlap then maybe
there's some quantum effect
but then in between that i i'm it's
probably more
action is more classical in terms of how
you pick things up it's literally like
i just keep doing it and in a very rough
and ready way i extract
some sense of how it is to pick up an
apple
or something you know and i'm inferring
things in a more classical way so there
that i think there's a mixture of a
classical and not only
past partially knowable set of
information transfer and maybe some more
quantifiable stuff which maybe is
i don't know if it gives you more
accuracy at that level but
um might be more quantum like but i i'm
not
that's that's my thoughts
um okay uh so just to dean the question
was
what's in between the model and the
inference okay

is that what you asked yeah so
let's just kind of go into that so what
by model
you mean the actual code as you wrote it
on the computer or the actual
systems diagram like the model you mean
what's on the computer or what's on the
paper is that what you
mean that's the product of the model
and then the inference is um the
computation that the model does or is it
is the
inference how we then look at that model
and then decide how we should do policy
because yeah it's it's literally in it's
it's moment
by moment it's that it's the smallest
time smallest of time frames relative to
the
the end the endgame or the end product
how do people
typically they put a plan in between
those two bookends
that's how they parameterize and so what
i'm curious is what other people think
actually is in there because i know we
see a plan
where we're given a plan but
i'm not sure that that's exactly what's
going on so i was just curious what
other people
thought is going on blue
so like i'm not sure that i'm
also like understanding the questions so
there's like the generative
model and the predictive model
right so there's these two models and so
the generative model

is like the model that you like have
um daniel probably have a better way to
say that and then the predictive model
is in my mind that's the plan like i
predict
that if i invest in crypto
i will have 10 more money at the end of
the year or whatever right like so
that's my prediction
that's my predictive model that's my my
my plan so in my mind
those two things are are synonymous but
but maybe daniel can
can clarify or speak to that in a better
way
thank you blue um i think i try to just
take what you wrote
annotated a little bit the internal
states is a generative
model of the niche and
that includes the agent's action in the
niche
like i can see myself going on a walk
every morning
that's like a combination of a
trajectory of action
and affordances and a niche and other
regularities and other
availabilities shoes or something like
that and then there's the actual
generative process
and the generic process is like the part
that is the play and also where
cybernetics theories like the quote good
regulator
theory come into play because the
cybernetics is like well you have to be
effective you have to have all the

variables

in your um environment but then we can
look at

for example alternate understandings of
astronomy

and there might be evolutionarily
adequate different understandings of
astronomy

that allow individuals to deal with
irregularities in their environments
for the niche that they're in given
their affordances

so the gen but the sort of realism take
would be that it's the same moon
and sun at the chart you know the
fibonacci spiral all these things you
can do

but it's about the narrative of policy
under that generative model the niche
but it's the same market that it gives
me an

ability to act in a non

a surprised way as opposed to using
surprise all

which is this kind of metric which i can
kind of use both in my retina and my
nerves and everything is this kind of
general awareness

of how much of a variation has been
but surprise and you know something can
happen to me once

and i make big decisions on that and
that is

sort of basically sitting on top of all
this activity sort of need to know

as an animal because and and um there's
a there's a great presentation on this
and i can't

at one point in history basically we
have to add in
who into our environment so up to then
it was just like
how apart and sometimes things fit
multiple categories so
how do we explain that with a plan no
because that's fluid
that's just dependent upon how you're
looking at it
and sometimes you'll you'll talk to the
person beside you and say do you see
what i see
which then you you've you've already
pointed that out a couple of times
in this conversation so i'm just trying
to figure out
what's in there is i think it's more
than a plan and i'd hate to default just
to that
or over reduce just to that because i
think that's part of the problem
when we're trying to figure this stuff
out
dean and um blue
so definitely there is a who and i think
that that really
like loops us back around to the point
of this uh
fields and glaze book paper that there's
this
intrinsic contextuality and and so you
can't really
effectively model contextuality
unless it's you can remove the intrinsic
component
right like so when you can when you can
remove the the subjective aspect like

i mean we had a conversation i don't
remember if it was last week or two
weeks ago about like
at a dinner party like how was the food
did you have a good time like
or not did you have like how was the
food was it fun and so all of this
is like you know there's no objective
answer

to these questions it's going to be
different for all the 15 people that
were at the dinner party right like
some person liked the food some person
had fun one person didn't one person
you know thought the food was cold or
didn't like the dessert or whatever
right there's

the subjective experience so it's really
difficult to separate

the who from contextuality

nice questions and uh answer stephen
yeah this is very this is very important
actually what you're saying

i mean i think this also ties into this
idea

of what it means when we do planning and
also how we think of

planning and start to extrapolate that
so generally speaking in the

in the everyday sense of planning we
have like some sort of shared

something in the niche like we have a
plan we often make plans

if not it's somehow there in some verbal
semantic thing which isn't just in my
head it is in the niche it's sort of in
this

semantic niche and but the the idea for

instance of
how does my body know the plan for
getting on a street car
right when the doors open right i i know
how to do that
um and it's not but that kind of plan
has to come from somewhere and it's much
more
there you know i mean so they get all
merged together but the
um and that maybe is partly in our niche
in the way that we have the right sort
of shoes on we
we set up the design you know because we
have some control
but then even when you get into other
areas of the niche okay i
come across a particular cliff in a
mountain
and i i'm you know i'm not from the
modern world so i haven't got all the
technology
right but you can reinterpret the past
right yeah that's uh not within our
affordances
it's we can only infer on certain things
but we have causal agency and actually
in the spm
textbook there are some integrals that
are like from
all of the time series together just
like sort of matches all the time
together
and there's other ones that are
calculated like up to that moment
in a sliding window and then it says
that's what makes it causal
is we've restricted the analysis to only

the previous states
that's kind of how mathematically
causality is defined as is like well if
it's lagged in time
or if you slide a window as the
arrow of time you know changes but then
that's not the same thing as the arrow
of time or the experience of time or
anything like that
so yep but then this local logic slide
was just to say that um
and to also continue this discussion on
contextuality
it's the local settings so how can i
where does this matter and how does this
expand the discussion
hopefully lead to more powerful models
is um
if you're not playing with the local
rules
then you're not playing the game and so
you're going to semantically lose
against nature if there's organization
in a way that attempts to violate
local logic like if there's some
equivalence between heat
and computation in our current
understanding
and you design a machine that just
violates those rules like you don't plan
to dissipate a certain amount of heat
given how much computation
it's like you're not designing within
the rules of the system you're just
making a poor computer
and so we're talking about communication
though and
information transfer that's meaningful

so

what are the local logics for
communication

what are the local logics that are
shared by people who speak the same
language versus all languages
versus all ages and things like that
what's the shared

context that lets us communicate what we
need or how we should act together
what else what would be a fun slide to
go to

or just topic to go to

it mentions ergodicity there that might
just be interesting to see his take on
that

yeah you know

blue do you remember but um i feel like
arrogancy wasn't

we can look actually in the paper itself
let's just see if it's used

there we go not used in the paper itself
but uh we put it in there for some
reason

partially related to these um kind of
what we're talking about now

in a way like as you're bringing up it's
relevant

um of a term but yeah it might be
there's a question about

it raises that question because it's
such a big part of active inference
how those two things relate

because it seems to be able to get at
some of the questions

without the need necessarily for
ergodicity in the same way

but maybe i'm not reading into it

um but i've got a feeling that it may
not require it because you have these
spaces
that maybe have inherent
ways of obtaining knowledge or
information between these different
types of
you know morphological transformations
okay i think here is where we asked it
so this is
the quote um in the paper
um they're talking about contextuality
by default
and they say the criterion effectively
generalizes
the intrinsic contextuality of quantum
theory
so the one that we know of with the wave
in the particle and the entanglement
spooky action stuff to the case in which
other properties of a context e.g the
order in which questions are asked
also affect the distribution of a
variable of interest
so it's like 20 questions
if you go down one branch first
it's going to be like a bifurcation with
a question and communication is like a
bifurcation
whereas in the telegraph in the shannon
info theory
it's like if a signal is garbled and it
doesn't come through it's like it's a
lost
transmission but semantically it it
might ruin the file
but at the level of the message it just
wasn't transmitted accurately

but um so it bifurcates the meaning of
the file

but from a signal to noise it didn't
have a categorical bifurcation
maybe only one bit in a million was
actually flipped
and that was just a stochastic thing but
when we think about semantic information
flow

the order in which the information flows
especially in a dialectical or in an
ecological context
is like the whole topic information flow
is improvised or at the very least it's
it's engaging

so it's not just letters on the page
re-capitulated but it's something
informational

stephen yeah that that
that's important and that ties into the
idea of

um how much there's like with shannon
entropy or there's the idea of a signal
there's something in the signal
and you're losing some of it now with um
kristen's work it's the idea but it's
it's it's all non-linear and you're you
need the action

so that you can compare um
the um the way that the the signal
changes are happening
to infer something from the noise so to
speak you need the action but in this
case

it's like there's another way to sort of
say there's an inherent
something out there so to speak there's
sort of something in the signal

um it's not just only the action needed
which is probably why then this can be
applied to bayesian inference and not
only active inference because
the saying there's something inherent
you don't only need the action and
you're getting it
because of the order in which the
questions are asked
um and that order
actually that's interesting then that
that isn't often how i thought of
contextuality i just think about
contextuality is where am i you know
what i mean and what's happening
but what order did what happened happen
give me something is new for me
no yep when are we
and when were we there all those
questions blue
so i have a comment on ergodicity but
can you back up
to the uh cone cocoon diagram because i
think that this is why we brought it up
yes so here
in this cone cocoon diagram the one on
the top
right like so so you see this um
information
channel c right that that relates all
these classifiers a1
a2 all the classifiers are linked to
this
core information channel and that that
kind of um
doesn't really imply like a a
distributed system
it applies like a centralized control

but
um we brought up ergodicity because
if if you look down in the second
picture here from c to d
there's this commuted commutative
property that is
or i guess it's even in the top picture
so it's this the g's right like the
commutative
um aspect of the cone cocoon
um diagram and so i think that that's
where where the ergodicity
um came into this paper
right like the fact that that all uh
properties are
available to all points in the in the
cone cocoon
cool and also you know these topics can
go
expand beyond the narrow and the
technical but
let's think about a technical definition
of sampling ergodicity so not taking on
all the
um experiential elements per se but a
remember is a classifier so it's a two
space
scarlet letter a that's what's being
represented with the
and it itself is a classifier
because when we were talking about the
two spaces there was like it could be
the um types of a computer language and
then what their descriptions are
so this is like cats are small you know
horses are big
this is my classifier you know big and
small and then horse and cat

and then just which one's big which one
is small so it's a two space
informally um and then here is
a different true space a_2 at a different
time maybe it's time point two
maybe it's person two maybe it's all the
people so here's me
a sub k all the k people or all the k
moments
and whether it's one person through time
if you ask the question you get the same
quantized response
yes the cat is smaller than the horse
because cat's small horse big
if you always get that response back
semantically
then you're sampling from ergodicity
with respect to
that question so there's it's a big
topic with like what it means in physics
and everything like that but just from a
sampling perspective if there's
 k people in the room and you sample in
all 100 of them
or every time you sample 70 out of 100
give you an answer
that's the sampling ergodicity and then
here are c and d which are like two
observers
and they're kind of co-sampling but they
can only see the raw data
so they're able to see all just what we
were getting at with the top part
just like this is like sampling across
informally population or through time
and then it's ergodic if there is a
stationarity there
here that is the same scenario but now

there's two observers
and then if they are related in this
very specific way
then it opens up a lot of connections
semantically between c
and d
also sarah added in the chat that
salience is a time dependent phenomena
and uh that's extremely true so
everything that we talk about with
attention
and salience and surprise those are like
almost explicitly time-bound
processes and even in machine learning
when people talk about oh it's a neural
network with attention
or with recurrence they're kind of
talking about a time
element but anything that's
where it's not just a snapshot that
moves state to state but there's
something that's captured across
time steps and that's called attention
sometimes so it's kind of like
it's actually a similar way to how it's
used in a modeling framework here
this was one question that we had in
0 and beyond and hopefully got to ask
chris a little bit but
like what
um are examples or how can we
you know what's the next question to ask
or what's the next
system to explore this way that was one
set of questions also if anyone in live
chat has any questions we can definitely
look at those
what else would be interesting i mean

there's a few more slides that we haven't looked at that we could kind of explore so maybe one element here is oh yeah go ahead steven sorry well just just once he talked about and maybe this is something he brought up as a as a slightly new advancement he talked about separability separability um as a sort of he positioned that sort of center stage during his slides when he did his introduction um as maybe a way to i don't know think about contextuality um in a more tractable way um yeah i wondered if that was you know was that how did that come up was that something oh that's interesting was that a little bit surprising that he when he did that um yeah great question and point so i just searched in the document there's two spots where it's mentioned they write we have previously shown how inference can be represented using the formalism of this paper employing only the quantum theory of separable systems and the thermodynamics of measurement interactions so there's the citation to fields and glazebrook to explore that more but i think from what we're talking about here's an interesting part that um is related to two spaces it is often the case

that two spaces may be separable or other types but let's just focus on the separable element um i.e both separable so it could be separable it's almost like you could have another two space you know which two spaces a through z are separable or not so there's separables equality and then it says separable means all rows are distinct extensional means all columns are distinct so actually the transform that goes between them is probably the transpose the t in r and then if it's separable on columns and you transpose it it's separable on rows so that's like a relationship that's kind of a high level but it's maintained when you transform it back and forth like blue is saying add subtract add subtract you can transpose the matrix back and forth um so separable meaning the rows are distinct so let's just think about the case of the chew space as a descriptor just which was one of the uh ways that they used it so like the types as types of processes kind of so if it's separable it's like they're distinguishable types sometimes that's also called a full rank matrix where it's like every row and column matters uniquely and informationally

there's no doubled rows
where you could condense the matrix by
saying actually there's five of this row
and so this or this row is a linear
combination of this other row
stephen yeah that
that um this this is good to bring it
back here that because it's uh
this is sort of the applications of it
and um the separability
um i think there's more that can be
looked at with this over time
i think this is obviously something
that's going to be quite
applicable and it also relates to you
know this idea of self-organization
and planned organization so you you know
how much are things um
uh able to to to
infer things because of the
um the separability um
that in itself um
maybe changes what's possible um
yeah actually here's a quote we haven't
looked at
this is really interesting though in
working set theoretically so
with this sort of set approach to choose
basis
there's considerable scope for the
choice of arguments in the choose space
as well as the choice of the
corresponding relation
r because we get to choose the spaces
and the map
so we can you know map from various
things to various things
different ways these and so what kinds

of

things can we do probabilistic in

particular conditional relationships

so that's like all of computation and

stats bayesian

fuzzy type relationships fuzzy logic

fuzzy math

um spatial observations

object identification and merological

reasoning this isn't a paper to look

into that means parts and holes

how they're related to each other and

dual process theories of cognition

so that's bringing it to the cognitive

neuro and to even the neuro scientific

numerous examples are discussed in the

paper and they are

it's um it's a double paper but it's

very informative

and so all these kinds of things are

kind of what

is adjacencies for the two space dean

and then anyone else dave if you have

any thoughts after dean otherwise no

worries

i just could you go back to that cocoon

thing

yeah that one there the bottom one it's

interesting because

when i read looked at that in the paper

it took me back to some

stuff i did about um that diamond

shape and how that symbolically relates

to opening up

which you mentioned daniel and and self

organization which stephen

mentioned um we see that when we see a

diamond

uh designating a high occupancy vehicle
lane
and the sort of way that that frees you
up as long as you
meet a minima in terms of the number of
people in your in your vehicle
and we used to talk about that idea
of how these spaces
are there even though you may not be
necessarily
noticing that um
yeah that's that's even if we don't
formally look at it that way
there's all sorts of examples around us
of when that's actually occurring
so i just thought i'd point that out it
may not have
necessarily do those kind of
associations but again if you're
trying to bring this kind of information
to people and go well here's an example
of
where we're actually doing this stuff we
may not be aware of it
but it's it's going on this that's
as soon as i saw that that's what kind
of made me smile
cool thanks for that association there
dave
this is not addressing any specific
point
but gregory bates and a number of times
observe the cyberneticians getting
stuck by talking about random inputs
random inputs he says look stop thinking
about random i want you to think about
noise we're now going to talk about
organization

from noise and i think part of the point
is
noise is more clearly relative to your
values and your plans
yes very nice and also affordances which
are
semantic like even years ago when i was
learning about shannon information
theory i thought i can be
symbolically surprised by the characters
on a page but if i don't speak the
language
who's to say whether the information is
really there so there has to be
something else here
that's helping address the situations
that are meaningful or not
because it's not just about how rare the
character is
in a world where just one word can mean
a lot
at the right time in the right context
stephen
yeah the the idea that you're
you're you you need to constrain certain
times
or separate i mean i suppose in a funny
way the car lane i
uh example is quite interesting is you
know you've got
the the movement of traffic and at some
point the high occupancy vehicle
lane becomes different in terms of
what's on it
as long as you can enforce a rule
oh and you know and communicate it but
and enforce it and that that kind of
idea that you can

create separability otherwise um
and maybe that that sort of principle of
uh there's some constraints um
implicitly out there i suppose on
what things are gonna do with each other
i suppose
um just like the way water behaves
in certain contexts at certain scales
within our gravitational field
but there's also and i'm not sure how
if i'm over extrapolating but how
biologically you know you know different
ways we have to
there has to be some point where you
have to impose a separability
you know you might need to form an organ
from
cells to be able to do the next thing
you know to
to create insulin or something it's not
just enough to have
some insulin cells dotted all around the
body you need to have a pancreas
you know so there needs to be that
constraint
you know so that you're in the high
occupancy lane and
that's what's happening so it's um
you're in the high insulin-producing
cell type
and then at a bigger level you're in the
high insulin-producing
organ sub-component in the organ and so
it's this nesting
and there's um uh maybe even an
extraction of
function using what you were talking
about earlier so then sarah wrote in the

chat also about the uh
definition of the two spaces here saying
that this is
another way in which the ergodicity is
going to come into play
because uh sarah writes separable is
saying that the event space for a given
time frame cannot be compressed
i'm not sure if that implies the
aggregate matrix itself cannot be
compressed reduced in some way
and then here this this by extensional
collapse
so if you can't collapse the rows and
you can't collapse the columns and you
can transform it and that still is going
to be like true it's just like it
it's informationally rich
then any repetitions in the rows of
objects and columns of attributes are
factored out
and also the reason why attributes are
at
time is an attribute if you think about
the object as existing through time
just like you could measure the color
every inch of the ruler
you could measure the length of the
ruler at every
measurement in 10 time points in
practice this removes unnecessary
repetitions in the content of
information
hence minimizes the amount of processing
required by a given algorithm
so if you do 100 measurements of exactly
one meter because the time was
unique that's still 100 measurements so

the classical approach would be like
well sounds like we got an answer i mean
look at the p-value it's one meter
but then here when we retain this rich
contextuality
we actually don't pool even though we
were sampling from ergodicity of length
we don't need to pool all of those
measurements together in the same way
perhaps
and so that is definitely related um
sarah's point that
this matrix the rows or the columns can
be thought of as being temporal
so when we think about the matrix having
structure on like a top to bottom or
left to right way
that can represent structure in time
like just like you have an eigenvector
you could have an eigenvalue
through time stephen so
if i'm understanding you you could have
a relatively
um say a small number of
uh observation types but build out
a space because over time you can still
you
the time variable gives you the ability
to
make that into some sort of space some
sort of topological space it's not
just a space of different measurements
of different things it's
sort of certain things over time
is that right yep here's an example that
kind of came to mind
um it relates to insect vision so a lot
of times

people think about insect vision like oh
they must see like a bunch of versions
of the world
but then we think okay well we have two
eyes and we see one version of the world
so it's probably
a little more integrated than buggy
putting that aside
what's the resolution of the bugs site
and it turns out that there's some
insects that only have like one
omittidia
cone so it's like kind of like a one
pixel
camera it's like almost light dark so
you would think
and if it were on the wall all it could
do is inference light dark in the room
but because the insect has a body is a
body
it can move so even if it's just light
dark or gradients of
it's getting like a bit string we can
think about continuous variables or
just zero one and so can a bit string
encode
information that's rich like yes like
sports casting
of a sports ball game you say now
they're running from the 30 and they're
at the 20
there's a person and they're at the 10
and so that's a string but it
conveys
action and context so they don't need to
take
the 4k video in order to know whether to
run

or walk and that's actually really related to

carl fristin's woodlice example which he claims is the origins of some of the free energy principle kind of

any other um slides we want to go to or we can just look at the last ones otherwise i think it's

been a great conversation and it was definitely a

educational paper and several weeks like for the lab we're always happy to have new participants

such as yourself dean so thanks a lot for joining

and yeah this was an interesting set of discussions also thanks a lot sarah for helping on

and and blue 4.0 for 17 because that that was kind of like how we started the conversation in a way in a way that helped us structure it because there's a lot to get lost in here blue what do you think and then dean

so you're on the markov blanket side can you go back to it yep

so i was really thinking about like i have this note in the paper

so hold on let me find it so in like you know when we have this markov blanket it's like the things that we're not privy to the information that we like don't necessarily

know about right and so we make a decision

we have all of the information that we need to make a decision but i was thinking like

in a system where the markov blanket is
like too big like if there's too many
things under the markov blanket that we
don't have access to like how does that
like

impact our um like ability to act
within the constraints of the system so
like

here it's where they talk about the
frame problem right like so
where um you know you kind of can't
solve a problem until it's been solved
already until you know how to solve the
problem like you can't determine the
answer and i think that that's like the
the gist of the frame problem feel free
to like elaborate or or correct me on
that

but um you know i think that that's like
one of the things that is so difficult
about a problem like

climate change right like so we can't
solve the problem like climate change
because we don't know

how many factors are potentially
influencing like we don't necessarily
have the entire context

for the climate change situation

so like i know what i'm doing i know
what my carbon output is

and and like i can research the
companies that i buy products from etc

but like i don't know what all of the
other nine billion agents in the world
are doing

and so like we can't you know make
predictions based on our actions

right so like is the markov blanket like

too big in that situation to
be able to solve the problem like
because we're not able to see
directly the uh outcome of our actions
or like even the actions of our group of
people that are in our network

i was thinking about that
interesting dean and then dave
um just a couple a couple of things
first of all um thank you for being so
welcoming

i really appreciate that and the second
thing is is that

[Music]

when i watched last week's um
broadcast because i watched it on
youtube i felt like i was
i was a spectator and now that i've been
a participant but i've been
a participant of a debriefing exercise
so i've actually been a participant
of something that's been separated out
from what originally happened
and i think that this paper is the same
this is real time what this paper was
discussing

um so if you want to i don't know if you
could find a better example of where
that idea of what perspective is and
what

potentially prisms are even if we're
not talking about them

we're still playing by their rules um
again thank you for allowing me to be
one of the

the people in the tile today i just
really really appreciate that
cool and it's like the paper is the tent

pole
in some ways because it's actually north
star the reference point but we're
giving our perspective
on it but we shouldn't be disagreeing
about what the paper is or says we're
looking at the same slides but then
the richness is actually everyone's
understanding and contribution just
wherever
it is coming from whenever it's coming
from there because what else could
that context provide dave
yeah i don't want to uh go too far off
track but i noticed the
paper references donald hoffman
and i don't want to do an injustice to
the good doctor
it may be that um i see him
stepping into it when he didn't really
with his one
video about the interface is
anywhere you want it to be it doesn't
make any difference
the interface is totally arbitrary maybe
that's just his billions and billions
moment but
did donald hoffman go down the rabbit
hole a couple of years ago or is he
still doing
good solid work that i can read and not
get my inexpert self thrown down a
rabbit hole
can't speak to uh the body of work don't
know it that much but just
i pulled up the citation from the fields
in glace brook paper
it's a 2015 paper of the interface

theory of perception
and so the way that they that fields and
glazebrook use it in the paper
is basically to um actually in a way
that's related
uh to the interfaces
it says equivalently an interface and
that's the citation so the markov
blanket
here is being connected to
their approach so it's kind of probably
a yes
and it probably is great insights about
the interface and then it probably does
matter
where it is or what it is or how it's
modeled so yeah nice
point though stephen
yeah the good good good observation
there i mean there's probably a you know
there's a reason why chris
fields has got involved in active
influence because
let's be honest as from my understanding
of my observations over the last 30 40
years consciousness and
physicist physicist physicists have kind
of run the show
around consciousness and there's been
more and more sort of
big claims um about uh you know
consciousness which is great and
um this is seen as um
you know and a quant there's a there's a
thing called um
mindell the mindells do the process
orientated psychology
and um some good work but i always

used to like have a bit of a sigh when
quantum got mentioned because i knew
that it was going to be
like even though i knew physics i just
felt and maybe i was wrong but i just
felt it was being used in a very
trans-personal
slightly hand-wavy way but maybe that
and
now we actually bring it into like
it's not all about the physics it's
about the biology
which is kind of like taking the
physicist physics
area down a peg or two um although now
kristen could say it takes up a peg or
two because it's all based on physics it
depends which way you want to look at it
but um
interesting world let's let's put it
like um thanks
yep in this quote um this is the authors
they're saying that this marco blanket
concept which is map the interface and
all these other ideas like communication
channels it's doing
a double role in these models
which is that it's where free energy is
minimized so it's sort of
what is going into the function that's
getting minimized if you're
you might be reducing your water usage
but what's going into your
actual function that's deciding your
policy that's what's getting minimized
and then something else might also be
affected
but what's getting minimized are the

blanket states and
it is that epistemic barrier that's
keeping
context what isn't directly measured
that information is hidden from the
observer so
there's probably a bunch of questions
about this and a lot to say about it but
one big question that i have is um we
know
from previous papers that one of
fristen's innovations
was to separate separability perhaps
separate
the blanket states into outgoing action
states in incoming sense states
now it's possible that there are certain
kinds of models that play
nice in the uh sort of undirected
bayesian
conditional relationship neutral
way because it's sort of like a built-in
bi-directional road it's sort of like
cars going both ways in the same lane
but when we really
make that separation clear and we really
have some states that have an
asymmetrical relationship in other
states that
have an asymmetrical relationship it
could be simpler
it could be more difficult so there
might be new
relationships that are enabled by the
separation there might be relationships
that are um put off
for now because it complicates it in a
different way

so i don't know what those would be but
those are just like what could happen
when you split the states or maybe
nothing happens
and it's already been dealt with but
yeah
and then sarah's uh perpetual slide on
the reservoir
computing um
we had a little oh the next steps were
just from their
words they had three areas and next
steps kind of we'll
go through this then we'll kind of give
anyone else give a last thought but
here's the areas where they're taking it
next or where they want to signal
for everyone to take it next so one is
exploring intrinsic contextuality in
humans it's like
short and sweet apply it to humans so
we're asking about applied active
inference about what is the applications
this paper
they are asking that too they don't need
to go into any more detail
the second question is about how context
switching is implemented
neurocognitively so like you're talking
to two people
and you're bilingual and they say
something in one language and the other
person says something a different
language
but semantically you're might not even
be
aware of that difference it might just
be associated with that person's voice

or with the
situation so how is that happening in
the brain or
in the body or different animals and
then the third question their broader
question so this is the real you know
that's the specula no
the third question is whether that
intrinsic contextuality
can be detected and characterized at
multiple scales in complex systems
so this is where it's kind of getting to
the collective intelligence or
distributed computational systems like
what if there were a signal that we
thought were uninformative but it were
actually reflective of this just immense
wisdom or processing power cognitive uh
dimension
so we might think that we're
communicating with all intelligent life
when we send out the golden
you know phonograph with the etching of
the fibonacci sequence or a tetrahedra
or our body or something like that
but there could be a whole dimension of
information that's just
at a resolution we're not even
perceiving
so that would be big of course how would
we detect that a priori with minimal
scale and system dependent assumptions
which are equivalent to throwing out the
noise and the signal together
in this way of thinking
well it would be great to learn more
about these
topics uh i hope that we can continue to

just have experts on
have all learners and participants on
because it's kind of
fun especially when we're talking about
stuff like this
so um any last yep steven
you're muted steven yeah just mention
you've got a bit of other
you know the next couple of events that
are coming up just so that people know
if they're interested um that you're
going to be talking with um
we're going to be doing the other live
stream oh yes
um let's look at the calendar
yep so we will
um yep we'll have 18
in the coming two weeks on the
predictive global neuronal workspace
with ryan smith and christopher white uh
hopefully joining for march 23rd and
30th but then also
march 18th in two days we'll have
demetrius bolas
so yeah that will be pretty cool
um thanks everyone for participating
great discussion so fun times